

Monolithically Integrated 2×226 Gbps Dual-Wavelength EML with a Passive MZI-based Optical Multiplexer

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Abstract We developed dual-wavelength integrated EMLs with a passive Mach-Zehnder optical multiplexer requiring no phase-tuning electrodes. The emission wavelengths are spaced for CWDM, and each supports 226 Gbps operation (113-Gbaud PAM4). Measured insertion losses were 1.8 dB at 1311 nm and 4.3 dB at 1331 nm. ©2026 The Author(s)

Introduction

In recent years, demand for higher capacity in data-communications systems has grown rapidly with the development of artificial intelligence and machine learning. Intensity modulation and direct detection (IM-DD) are typically used for short-reach links (<10 km). 100 Gbaud PAM4 optics are expected to be deployed for 800 GbE and 1.6 TbE links [1]. The electro-absorption modulated laser (EML) is a promising solution for these applications because of its simple configuration, small footprint, high bandwidth, high extinction ratio, and low drive-voltage requirements. Recently, several studies have demonstrated high-speed operation exceeding 200 Gbps by driving EMLs at rates above 100 Gbaud using pulse-amplitude modulation formats such as PAM4, PAM6, and PAM8 [2-4]. While these demonstrations underscore the technical potential of EMLs for high-capacity links, widespread deployment will depend not only on device performance but also on system- and manufacturing-level considerations. Demand for optical transceivers is expected to continue rising, making it critical to address practical challenges such as overall transceiver miniaturization and reduction of manufacturing costs. Conventional architectures that rely on external optical multiplexer (MUX) such as prisms coated by thin-film filters increase the footprint of optical components and raise assembly costs due to the requirement for precise alignment.

To address this issue, we have reported a dual-wavelength integrated EML chip for 226 Gbps/λ PAM4 driving, in which two EMLs are combined via an integrated optical MUX to emit two wavelengths through a single output port [5]. This device is expected to reduce the transceiver footprint and to contribute to lower manufacturing costs by eliminating external optical elements. However, in our previous work we used a 2×1 multimode-interference (MMI) coupler as the on-chip MUX to combine light incident on two input ports into a single output port. This MMI configuration inherently incurred a 3-dB insertion loss, which substantially reduced the optical output power. To resolve this issue, we incorporated a Mach-Zehnder interferometer (MZI)-based MUX, which functions as a theoretically lossless multiplexer, thereby avoiding the optical-output power penalty. Furthermore, by appropriately adjusting the arm waveguide lengths, the device can be implemented as a passive configuration that eliminates additional phase-tuning electrodes.

Design of Dual-Wavelength EML

Figure 1 illustrates the schematic structure of the developed dual-wavelength EML chip. Each EML features a unique hybrid waveguide structure, monolithically integrating a buried heterostructure (BH) distributed feedback laser diode (DFB-LD) for improving heat dissipation and a high-mesa electro-absorption modulator

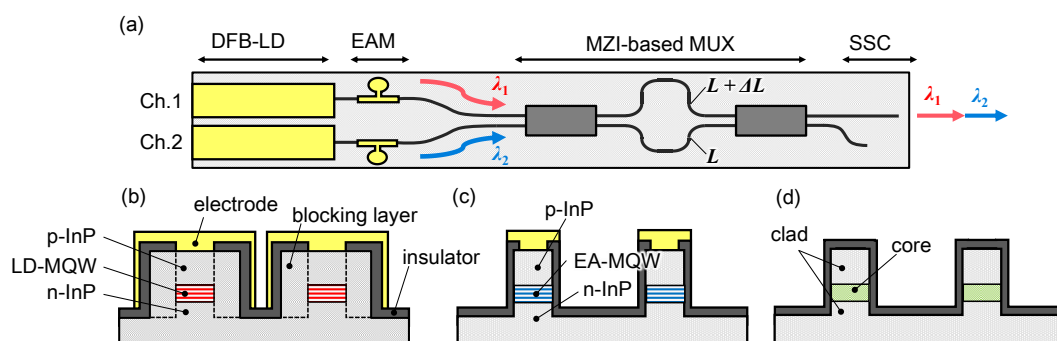


Fig. 1: Structure of monolithically integrated dual-wavelength EML (a) top view, (b) cross-sectional view of DFB-LD, (c) EAM, (d) arm-waveguide of MZI-based MUX

(EAM) waveguide for lowering capacitance and higher optical confinement due to its larger horizontal refractive index step compared to buried or ridge-type waveguides [6]. Two sets of these DFB-LD and EAM were integrated, with the emission wavelength for Ch.1 set to 1331 nm and for Ch.2 set to 1311 nm.

To combine these two beams, an MZI-type MUX was implemented, comprising an input-stage 2×2 multimode-interference (MMI) coupler, two single-mode arm waveguides, and an output-stage 2×2 MMI coupler. The 2×2 MMI coupler is based on paired interference, and its length was set to half the beat length, $L_\pi/2$ [7]. A path-length difference ΔL is introduced between the two arm waveguides in their straight sections. By matching ΔL to the beat length of the input wavelengths λ_1 and λ_2 , both wavelengths can be routed to a single output port. In the experiments, ΔL was varied in fine steps of 50 nm from 10.0 μm to 10.6 μm to fabricate a series of samples. A high-mesa structure was employed in the MUX region to suppress radiation loss in the bent waveguides. Of the MUX's output ports, one is the main waveguide equipped with a spot-size converter (SSC) and a window structure to improve fiber-coupling efficiency. The other output port is implemented as a dummy waveguide to prevent interference with light emitted from the main waveguide.

Characteristics of MZI-based MUX

Figure 2 shows the measured optical outputs of the two channels obtained with each DFB laser diode driven at 100 mA under the case temperature of 55°C. As shown, the optical output at both wavelengths varies sinusoidally with ΔL , with a period of approximately 0.4 μm . This period agrees well with the value calculated from the emission wavelength and the effective refractive index of the high-mesa waveguide ($n_{\text{eff}} = 3.22$), indicating that the MZI-based MUX reproduces the expected behavior. In addition, both wavelengths exhibit peak output near $\Delta L = 10.2 \mu\text{m}$, confirming that this device design successfully

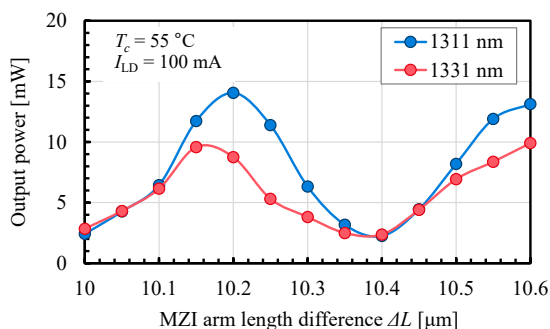


Fig. 2: Optical output power of Ch1 ($\lambda = 1331 \text{ nm}$) and Ch2 ($\lambda = 1311 \text{ nm}$) as the MZI arm-length is varied.

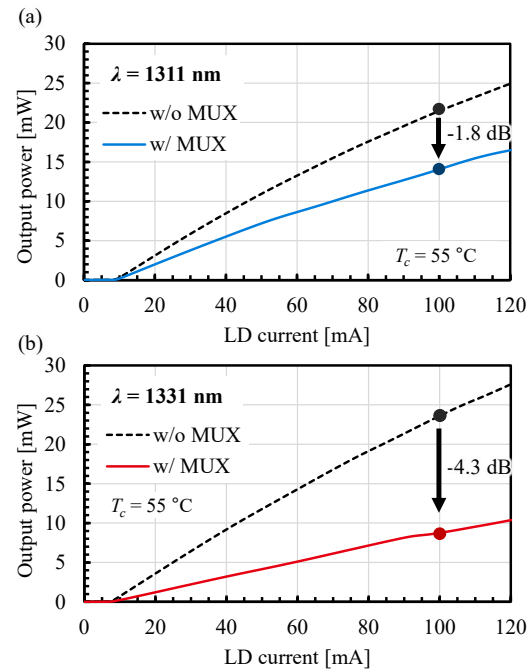


Fig. 3: P-I characteristics of (a) Ch2 (1311 nm) and (b) Ch1 (1331 nm), with and without the MUX.

multiplexes the two wavelengths into a single waveguide.

Figure 3 shows the P-I characteristics measured under CW operation at 55°C for samples with and without the MZI-based MUX. The data for the sample with the MUX correspond to the device with ΔL of 10.2 μm shown in Figure 2. Compared to the device without the MUX, this device shows a reduced slope efficiency, which is attributed to the MZI-based MUX insertion loss. The insertion losses were calculated as 1.8 dB at the wavelength of 1311 nm and 4.3 dB at 1331 nm. The observed wavelength dependence of the loss is attributed to the difference of insertion losses of the 2×2 MMI. This issue will be discussed in detail in the next section.

Next, we also performed PAM4 evaluations at 113 Gbaud. An arbitrary waveform generator (AWG, Keysight M8199B) generated an electrical PAM4 short stress pattern random quaternary (SSPRQ) signal, which was pulse-shaped by a raised-cosine filter. Its peak-to-peak modulation amplitude (V_{pp}) was set to 1.3 V. The optical output from the EML was coupled to a sampling oscilloscope (Keysight N1032A) in a back-to-back configuration. The received signal was equalized using a 15-tap T-spaced feed-forward equalizer (FFE), and the TDECQ was subsequently measured, which were calculated at a target symbol error ratio (SER) of 9.6×10^{-3} . Figure 4 presents the measured optical waveforms. No degradation was observed even when the MZI-MUX was

integrated. The samples without and with the MZI-MUX showed the extinction ratios (ERs) of 6.7 dB and 6.4 dB, and TDECQ of 3.3 dB and 3.0 dB, respectively.

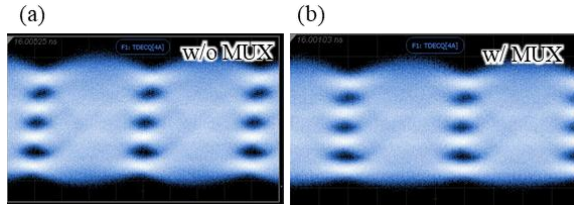


Fig. 4: Optical waveforms after equalization for the samples (a) without and (b) with the MZI-based MUX under 113 Gbaud PAM4 modulation.

Analysis of Loss Components

To analyze the insertion loss of the MZI-based MUX, the loss was separated into propagation loss in the arms and the insertion loss of the MMIs.

First, to evaluate the propagation loss of the arm sections that include bent waveguides, two types of samples were prepared: one in which the MZI-based MUX region shown in Fig. 1 was replaced with straight waveguides only, and another that incorporated four 90° bent waveguides and the straight waveguide segments between them. The 90° bends were implemented using clothoid curves to reduce transition losses caused by mode mismatch. Figure 5 plots the propagation loss derived from the difference in optical output between the two sample types. The horizontal axis is the straight-waveguide length ΔL , and the vertical axis shows loss values that include propagation loss in the straight sections and radiation loss from the bent waveguides. Error bars denote one standard deviation ($\pm\sigma$). As shown, varying ΔL does not change the loss beyond the error bars, indicating no significant propagation loss in the straight waveguides over the investigated length range. Hence, the measured loss is attributed primarily to radiation loss in the bent waveguides, which is estimated to be approximately 0.6 dB.

Next, we evaluated the insertion loss of the MMI. The MMIs that form the MZI-based MUX in Fig. 1 are 3-dB couplers with length $L_\pi/2$ (half the beat length); for this evaluation, samples with twice that length (L_π) were fabricated and the optical output after transmission through the MMI was measured. If the MMI length exactly matches the beat length, it should be lossless in principle, and all optical power should be coupled into the bar port. Figure 6 shows the optical output at 55°C, 100 mA for samples with and without the MMI, measured at wavelengths of 1311 nm and 1331 nm. The insertion loss of the MMI was 2.6 dB at 1311 nm and 3.5 dB at 1331 nm. These relatively large insertion losses are attributed to a deviation of the MMI length from the exact beat

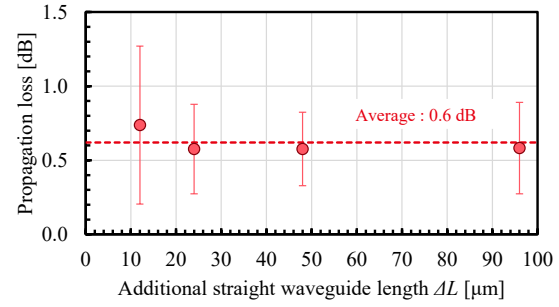


Fig. 5: Propagation loss of the MZI arm, including bent and straight waveguides, as a function of straight-waveguide length

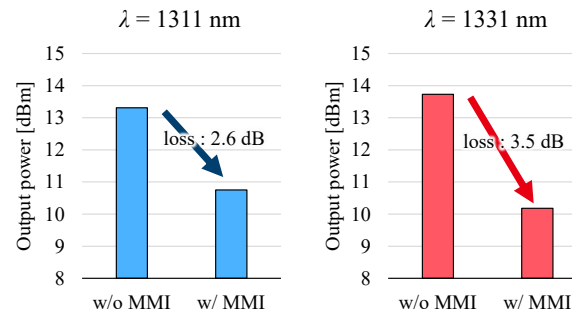


Fig. 6: Optical output for samples with and without MMI at wavelengths of 1311 nm and 1331 nm.

length, which causes the output ports to be displaced from the imaging points. In addition, the beat length differs by several micrometers between these two wavelengths, which likely results in the higher loss observed at 1331 nm. This result is also consistent with the previous chapter, which reported that the insertion loss of the MZI-based MUX was larger at 1331 nm than at 1311 nm. However, it should be noted that the loss evaluated here was measured using MMIs of twice the length of those included in the MZI-based MUX; therefore, the loss measured in this evaluation does not directly correspond to the loss of the MMIs within the MZI-based MUX.

Summarizing the breakdown of losses for the MZI-based MUX, the radiation loss of the arm waveguides is approximately 0.6 dB, and most of the MUX loss is attributed to the insertion loss of the 2×2 MMIs. It could be reduced by measures such as precisely matching the MMI length to the beat length and optimizing the input/output port geometries to minimize coupling loss; these improvements remain tasks for future work.

Conclusions

We reported an InP-based 2 × 226 Gbps dual-wavelength integrated EMLs with a passive MZI optical multiplexer requiring no phase-tuning electrodes. These results are expected to contribute to the realization of high-density, low-cost optical transceiver and to support the broader expansion of optical communications.

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